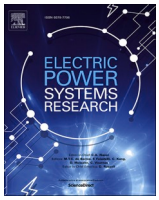
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A new adaptive protection approach for overcurrent relays using novel nonstandard current-voltage characteristics

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ABSTRACT

Due to the high penetration of renewable energy sources in the electrical power network, tripping time and ideal overcurrent coordination are currently among the key power system protection problems. In this paper, a new adaptive and optimal coordination approach for the Over Current Relays (OCRs) in a power network coupled to a Photovoltaic System (PV) has been devised employing nonstandard voltage-current tripping characteristics. This research presents and proves the novelty of the proposed adaptive optimal coordination method based on Genetic Algorithm (GA) to achieve the minimum total tripping time compared to conventional and nonstandard current characteristics used in the literature. The proposed adaptive optimal coordination strategy uses the nonstandard voltage-current tripping characteristics to minimize the operational time of OCRs in an effort to increase the sensitivity and reliability of the protective system. Results of the proposed optimal coordination scheme are compared to conventional characteristics and nonstandard current characteristics in the literature using a case study involving an IEEE-9 and IEEE-33 bus systems connected to two PV plants. This case study was developed and presented using industrial software (ETAP). Utilizing various fault and power network model scenarios, the suggested nonstandard voltage-current OCR coordination method and optimization algorithm are assessed. The findings in this paper show that the suggested nonstandard voltage-current optimal nonstandard scheme successfully decreases the total running time of OCRs and enhances relay sensitivity under minimum and maximum fault scenarios. The proposed adaptive optimal coordination approach reduces the overall operational time of all OCRs compared to the conventional and current characteristics of the literature by 57.9% and 54.2%, respectively.

1. Introduction

1.1. Background

The rise in electrical demand due to the electrification process for vehicles, increase of population and limitation and prices of fuel, are occurring globally. Researchers have focused in using renewable energy resources as effective and alternative energy solution in response to these concerns about the economic and environmental effects of current energy sources [1]. Distributed Generation (DG) based on renewable energy resources such as PV and wind systems are expected to be more frequently integrated into the Distribution Network (DN) as a significant replacement for traditional energy sources [1,2]. However, the DN's face complex challenges as a result of the rising penetration of DG units into the electrical grid. For example, DGs create power protection problems due to the new and stochastic nature of faults and the topology of the power network. Traditional DNs have been transformed by integrating DGs from radial into non-radial and multiloop networks. Modern DNs with stochastic bidirectional power flows lead to wide range of fault currents and directions compared to traditional network's [1,2]. In addition, DNs can be operated based on the islanding mode, which increases the complexity of designing and developing selective and sensitive protection systems. As a result, one of the most complex problems in the management and planning of power systems is choosing the optimal protection scheme [2–4]. Therefore, the significant role of developing adaptive and optimal coordination configurations is to ensure network security and reliability while achieving the shortest operational time for system relays such as the Over Current Relay (OCR).

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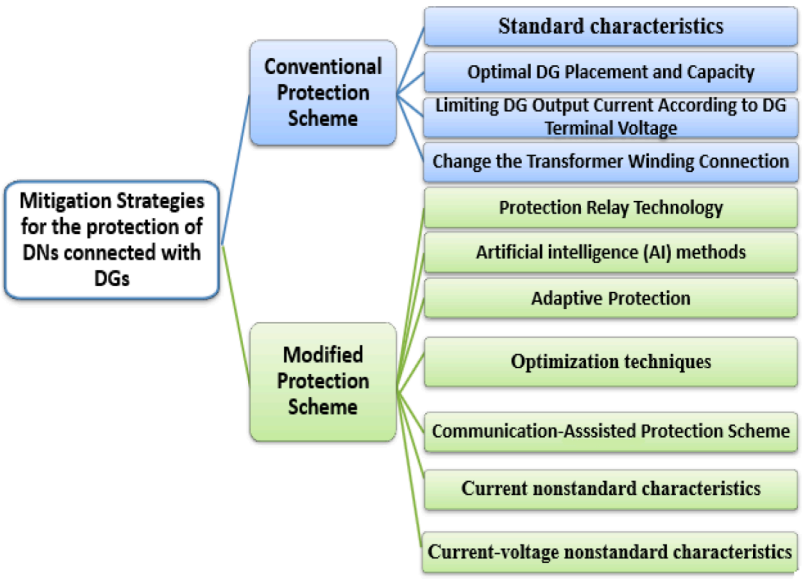


Fig. 1. The OCR protection approaches for DNs equipped with DGs.

This article aims to fill this gap and develop an innovative adaptive and optimal protection approach for OCRs coordination using a new nonstandard current-voltage characteristics to guarantee a DN reliability and achieve an OCR minimum operational time.

1.2. Literature review

Many researchers have thoroughly investigated power system protection problems and proposed different solutions to obtain optimal coordination and fast trappings using conventional OCRs characteristics [5]. Commonly, the most practiced conventional OCRs types (standard characteristics) bare definite-time, inverse-time, instantaneous, and mixed curves characteristics. These approaches that depend on standard curves, such as the inverse-time scheme, are common for traditional networks without DGs; however, they show limited ability to deal with challenges of modern DNs with DGs due to the stochastic nature of the load and generation profiles and different power flow and fault scenarios [6]. In the literature, there is a limited focus on developing innovative protection approaches for OCR arrangements for modern DNs with DGs, the PV as an example. One of the most commonly proposed solutions for addressing the coordination of protection schemes in modern DNs is to formulate the protection problem as an optimization problem. Then, using different optimization techniques, the optimum coordination for the protection schemes is obtained [7,8]. The authors in [9] solved the OCR coordination problem as an objective function under certain constraints using modern metaheuristic algorithms, namely: teaching learning, grey wolf optimizer, moth-Flame and particle swarm optimization algorithms. The optimization techniques improved the performance of the OCRs and minimized the overall operational time with a suitable characteristic curve. Mohammadi et al. [10] and Adelina et al. [11] used a Genetic Algorithm (GA) to find the optimal OCR settings for the primary and backup relays. However, their proposed characteristic curves of OCR do not operate under different fault locations and types and faced challenges to provide fast response with stochastic and volatile nature of load and generation profiles. In this article, the proposed adaptive protection approach will be tested on two power networks (IEEE-9 and 33 bus) under fault scenarios to test the sensitivity and selectivity compare to traditional approaches from the literature. Yazdaninejadi et al. [5] and Ezzeddine et al. [6] developed optimal coordination approaches by having more flexibility in adjusting the directional OCR characteristic [5] or optimizing the value of the pickup current and time multiplier setting (TMS) for directional OCR [6]. The results of [5] (by using IEEE 8 and 14 bus systems) and [6] (on IEEE 8 and 30 bus) system did not investigate the impact of DGs on the network or the performance of the protection system. The proposed approach in this work depends on two factors: current and voltage to improve the protection system performance in power networks with DGs compared to the conventional and nonstandard characteristics.

Recently, many researches proposed the use of nonstandard characteristics for dependable system protections that are capable of responding to modern DN challenges [1,9–13]. Kiliçkiran et al. [13] introduced in his review paper the significant use of utilization of nonstandard characteristic in power protection system and addressed the gap of developing new adaptive approaches to solve relay coordination problems. The outcomes of the non-standard (logarithmic function) coordination scheme were compared to conventional OCRs characteristics and showed better performance in terms of operational time and relay sensitivity [1,9]. Other authors, as [10] used the measurement of fault levels and locations, while the authors in [11] modified the constants of the time characteristic curves to adapt and create nonstandard tripping characteristics. However, previous studies [1, 9–13] depended only on current time characteristics to achieve the optimal operational time for OCR. Therefore, the authors in [14,15] presented new time–current-voltage tripping characteristics that can provide an achievable reduction in total time. The time–current-voltage tripping characteristics in [14,15] are formulated as a constrained nonlinear programmable (NLP) problem based on standard OCR characteristics. The proposed solution in [14,15] which was based on a standard characteristic curve does not consider the absence of high integration of DGs under different fault nature, magnitude, and pickup current. The OCR protection approach for DNs equipped with DGs is mainly divided into conventional and modified scheme, as shown in Fig. 1. In general, the speed of standard characteristics is not a high rapid speed in terms of tripping relay time and total operational time. Standard curves depend only on the current-time characteristic and are suitable for phase faults more than ground faults. On the other hand, nonstandard characteristics consider the increment of DGs in DNs by mainly employing only the current-time characteristics. However, the current nonstandard approach needs more improvement to achieve the coordination time interval constraints and to minimize total operational time. Therefore, this work presents an innovative adaptive protection approach using nonstandard current-voltage characteristics to achieve optimal operation time for the OCR under multi-fault and network operation scenarios.

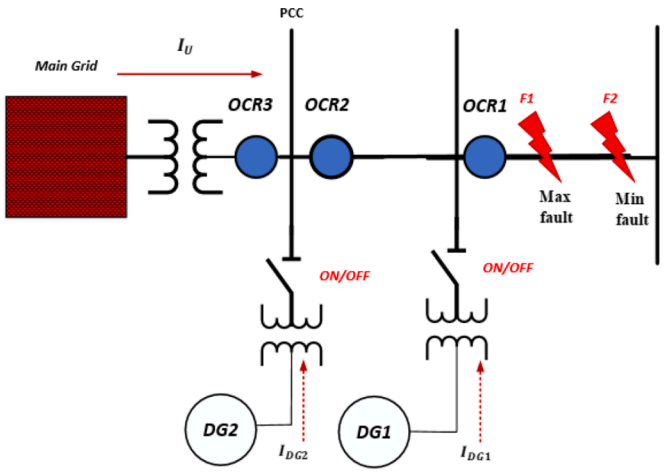


Fig. 2. Single line diagram of DN with two DGs and three OCRs.

1.3. Contributions

The modern DNs are more dynamical than traditional power grids and require high flexible and advance power protection schemes. The aforementioned evidence shows the need for OCR protection schemes that are compatible with dynamic power flows, with fault and network topology to overcome the raised questions about the appropriateness of standard and nonstandard OCR protection schemes (in terms of stability and operational time). Therefore, this work aims to presents a new adaptive nonstandard current-voltage tripping characteristic to deal with the different fault levels and achieve optimal operational time under multi scenarios of network operation compared to conventional and nonstandard current characteristics from the literature. The proposed adaptive and optimal approach based on the new nonstandard time current -voltage characteristic will be solved by the GA method to achieve minimum operational time for the OCR and to improve the protection coordination performance. The key contributions of this paper are presented as follows:

i Developing a new adaptive nonstandard current-voltage tripping characteristic that incorporates and deal with the impact of stochastic DGs behavior and the varied level of fault current.

ii Developing an innovative optimal OCR coordination approach using new nonstandard current-voltage characteristics to achieve minimum total tripping time compared to conventional and nonstandard current characteristics used in the literature. A substantial decrease in the overall total operational time is targeted to be achieved with no record of miscoordination.

iii Finally, testing and evaluating the sensitivity and performance of the proposed new scheme under different fault scenarios and multi DN operation scenarios considering the islanded operation mode. This evaluation aims to show the capability of the new adaptive nonstandard current-voltage tripping scheme to accommodate and handle different DN operational modes.

1.4. Outline of the paper

The rest of this article is divided into the following sections: Problem statement and solution plan are presented in Section 2. Then Section 3 illustrates the methodology of the novel optimal OCR protection scheme and the new time current-voltage characteristic. Section 4 discusses the results of the analysis and simulation. Section 5 presents the summary and new findings of this work.

2. Power protection challenges: illustration-based analysis

To provide reliable and safe working conditions, protection of the DN equipped with DG, such as PV systems, is essential [14–17]. The following example introduces the power protection challenges in modern DNs. Fig. 2 presents a single line diagram of a DN equipped with two DGs. The network is protected by three OCRs (OCR1, OCR2 and OCR3) which have been coordinated from the load side to the main grid source side. In case the maximum fault (F1) and minimum fault (F2) are occurred, as shown in Fig. 2, the OCR1 and OCR2 will react as primary and backup relays, reactively. In general, the Coordination Time Interval (CTI) between the relay's ranges between 0.2 and 0.5 second, which leads a large total operational time of OCRs. Integrating of the two DGs in this DN will change the fault characteristics, such as values and fault resistance, the network topology, loading/generation level variations, and islanding operation mode. These changes in the power flow and fault levels will lead to a miscoordination event at the OCRs system. Fig. 3 shows an example of miscoordination in standard protection curves. If OCR1 fails to operate at a minimum current fault (F2), OCR1 will not operate at this new level of fault current, as shown in Fig. 3. As a

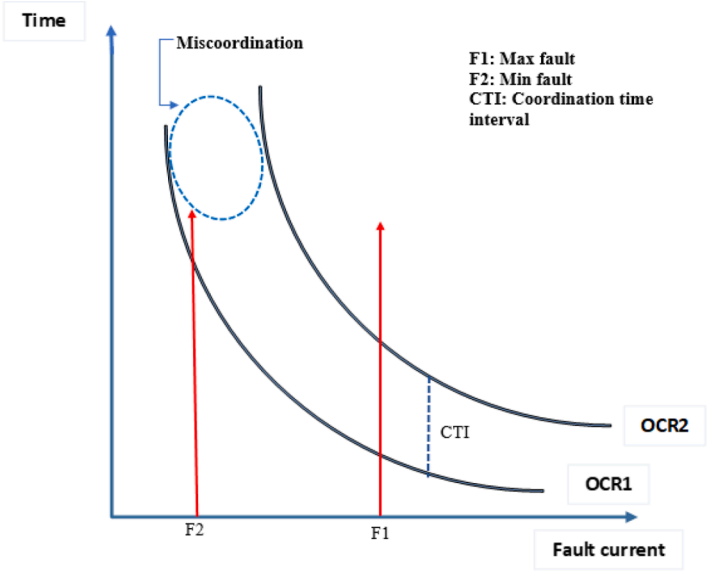


Fig. 3. Standard characteristics and miscoordination between primary relay and secondary relay in presence of DG.

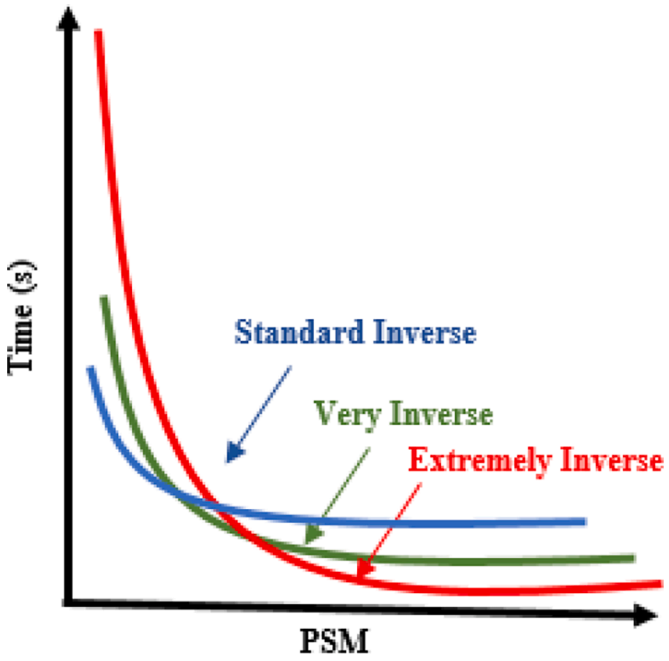


Fig. 4. Typical standard OCRs characteristics.

result, the standard protection plan and coordination will be ineffective in dealing with the wide range of minimum faults in the network with DGs. Standard characteristic curves and coordination settings will not be able to obtain the required selectivity, sensitivity, and speed terms for system protection [1,14–17]. To overcome the miscoordination problem and power protection challenges in modern DN, this article designs and develops a new and intelligent OCRs coordination scheme. In order to achieve a minimum operational time for the OCRs without miscoordination events, the Plug Setting Multiplier (PSM), Time Multiplier Setting (TMS), and CTI for the coordination curves need to be optimally determined.

2.1. OCRs coordination based on standard characteristics

The conventional coordination between the OCRs is primarily decided in the typical protection scheme under the assumption that the network conditions and parameters will not change during a fault. Eq. (1) describes the CTI between a primary relay and a backup relay, in F2, as shown in Fig. 2 [1,16].

$$t_{OCR2} - t_{OCR1} \geq \text{CTI} \tag{1}$$

where t_{OCR2} and t_{OCR1} are the tripping times for OCR2 and OCR1 respectively. As previously discussed, volatile power flow and fault nature will require a CTI that is less than the traditional CTI for conventional OCRs which causes miscoordination events. The conventional OCR commonly works under the Inverse-time curve, which is inversely proportional to the fault current. As a result, with extremely high currents, the OCR will trip quickly. According to the IEC 60255, Eq. (2) describes the operating time of the inverse-time relay, t, [1]:

$$t = \left[\frac{k\beta}{(PSM)^{\alpha} - 1}\right] \times \text{TMS} \tag{2}$$

$$PSM = \frac{I}{I_{pickup}} \tag{3}$$

where β and α define the slope of the curve characteristic, k is the time multiplier, I is the fault current, I_{pickup} is the pick-up current, PSM is the plug setting multiplier, and TMS is the time multiplier setting. The parameters β and α will be determined based on the type of inverse, very inverse curve characteristics, and extremely inverse, as shown in Fig. 4 and Table 1 under the IEC 60255 standard. In general, the inverse time-current characteristics are frequently utilized in the conventional protection system. The inverse time-current properties are typically restricted to particular low and high PSM values [1,18]. The modern DNs with DGs have challenges deal with different fault characteristics when using the conventional protection system. For example, the varied values of maximum and minimum fault currents will restrict employing the inverse time-current characteristics.

Table 1
Standardized Characteristics for OCRs curve.

Relay type	α	β
Inverse	0.02	0.14
Very inverse	1.00	13.50
Extremely inverse	2.00	80.00

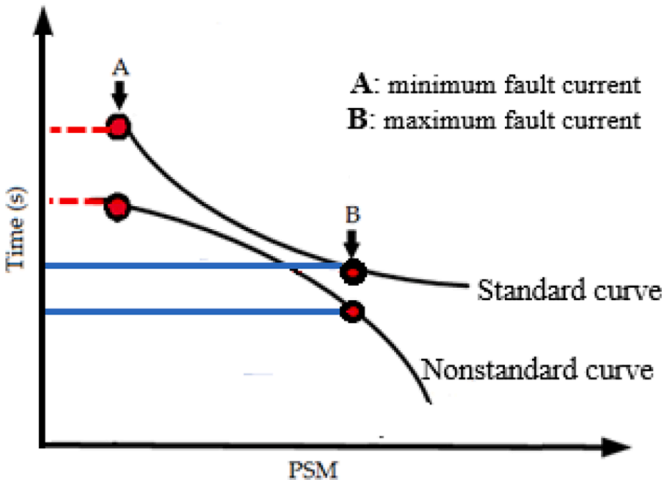


Fig. 5. A comparison between standard and nonstandard curves in terms of tripping time.

2.2. OCRs coordination based on nonstandard characteristics

A complex OCRs coordination strategy using a nonstandard time tripping characteristic based on the logarithmic function is described and presented in [1,18]. Addresses the limitations of using the inverse time-current characteristics in modern DNs. The authors in [1,18] recommend using nonstandard time characteristic curve compared to standard characteristics where it shows higher performance. In contrast to the conventional characteristic curve, the nonstandard characteristic in [18] offers a lower operation time and covers the minimal fault region for the back relay, as seen in Figs. 4 and 5. By reducing the duration of tripping in the cases of minimal and maximum faults, the nonstandard characteristic curve in [1,18] avoids the OCR miscoordination issue in the power network with DG. The curve is also used in a wide range of applications including problems with thermal stress in machinery such as transformers and cables. The nonstandard tripping property and the tripping time, t, is described by Eq. (4) as follows:

$$t = \left(5.8 - 1.35 \times \log_e\left(\frac{I}{I_{pickup}}\right)\right) \times \text{TMS} \tag{4}$$

The proposed nonstandard time–current in [1,18], as described in Eq. (4) and as shown in Fig. 5, was developed based on the logarithmic function, $\log_e$, and constant coefficients for all OCRs. Therefore, the operational time will be less dependent on the fault characteristics, which improves the performance of the protection system in terms of

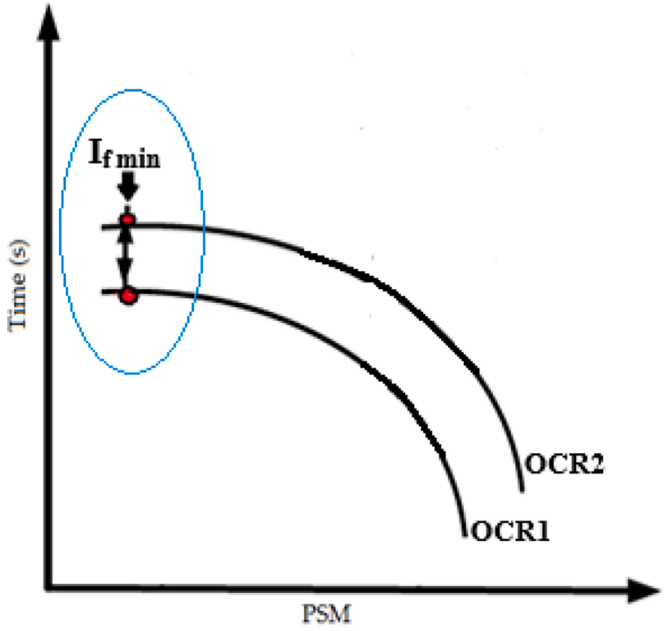


Fig. 6. Nonstandard current tripping characteristics for OCRs cover the minimum fault area.

selectivity compared to standard curves, as shown in Fig. 5. In addition, the proposed nonstandard time-current tripping characteristic type covers the minimum fault area and avoids miscoordination events, as shown in Fig. 6. However, there are some limitations to using the proposed nonstandard time-current tripping characteristic type, which are summarized as follows:

- Dependable only on the magnitude of the fault current magnitude.
- The curve is fixed and not adaptive.

These limitations reduce the ability of achieving the minimum operational time for different DN operation conditions and different fault levels. Therefore, the proposed work in this paper aims to design and develop an innovative protection approach using nonstandard current-voltage characteristics to achieve the optimal operational time for the OCR in multi-fault and network operation scenarios compared to the standard and nonstandard tripping characteristics in the literature.

3. Proposed method: nonstandard current-voltage characteristic

The limitation of employing standard (inverse curve) and the nonstandard (time-current tripping) characteristics will minimize the sensitivity of OCRs and achieve minimum tripping time. To overcome this limitation, an advanced OCRs coordination scheme based nonstandard current-voltage characteristics is presented in this section. This work presents and compares this advanced protection scheme to the standard and nonstandard time-current tripping characteristics. Generally, numerical OCRs have the ability to modify relay operating characteristics using online data. In addition, numerical OCRs allow the use of standard characteristics or create new tripping characteristics to deal with the challenges of modern DN [18–22].

3.1. A nonstandard current-voltage characteristic

The standard and time-current characteristic depends on the fault current values where it reduces the tripping time as the fault location becomes closer to the source (maximum fault current) [18,20,23], as shown in Figs. 3 and 6. In addition to that, these two curves have limitations, as previously discussed, which lead to low-speed tripping in the

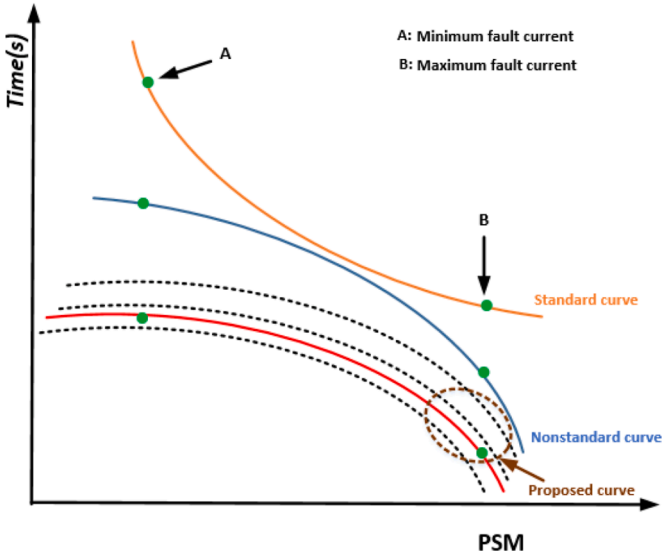


Fig. 7. A comparison between the proposed characteristic and standard and nonstandard curves.

DN with DGs. The fundamental reason for this is that the fixed characteristic curves of standard and nonstandard schemes [1,18] will not be able to handle the high penetration of DGs and their multiple changing operation modes. In this work, the proposed nonstandard current-voltage curve is used to obtain an adaptive characteristic under different fault locations and types which leads to achieving a minimum tripping time for the OCRs and avoid any miscoordination records. The nonstandard current-voltage characteristic is described by the following equation:

$$t = \begin{cases} \left(5.8 - 1.35 \times \Delta V \times \log_e\left(\frac{I_{sc}}{I_p}\right)\right) \times TMS & \Delta V > 1 \\ \left(5.8 - 1.35 \times \log_e\left(\frac{I_{sc}}{I_p}\right)\right) \times \Delta V \times TMS & \Delta V < 1 \end{cases} \quad (5)$$

where the operational tripping time, t, for the relay is determined based on the bus voltage per unit, ΔV, and the logarithmic function, $\log_e$. Eq. (5) presents the proposed nonstandard current-voltage characteristic for all OCRs utilizing logarithmic, the bus voltage and constant coefficients.

- Firstly, the voltage term, ΔV, is used to create an adaptive characteristic that can be modified based on instantaneous voltage and operation mode, as shown in Fig. 7. This will increase the selectivity of the protection system without affecting fault currents or locations.
- Secondly, the grading time must be consistent and unaffected by the location of the fault in the network or the fault current level to guarantee selectivity in the OCRs coordination.
- Finally, the nonstandard term, $\log_e$, will help in covering the islanding operation mode and minimum fault level. In general, finding the minimum faults was typically difficult for the normal inverse curves. To ensure selectivity without any miscoordination event, the suggested nonstandard current-voltage tripping characteristic, as shown in Fig. 7, provides a sufficient region for detecting and coordinating the OCRs during minimum faults.

In this paper, the proposed new adaptive protection approach will be examined and compared with the standard and time-current tripping characteristics. The new strategy attempts to address the coordination issue for the OCRs and to reduce the overall operational time in the DN with the DGs. To provide an adaptable solution with the best coordination of OCRs, this novel methodology will combine both current-time characteristics and voltage characteristics. A comparison of standard

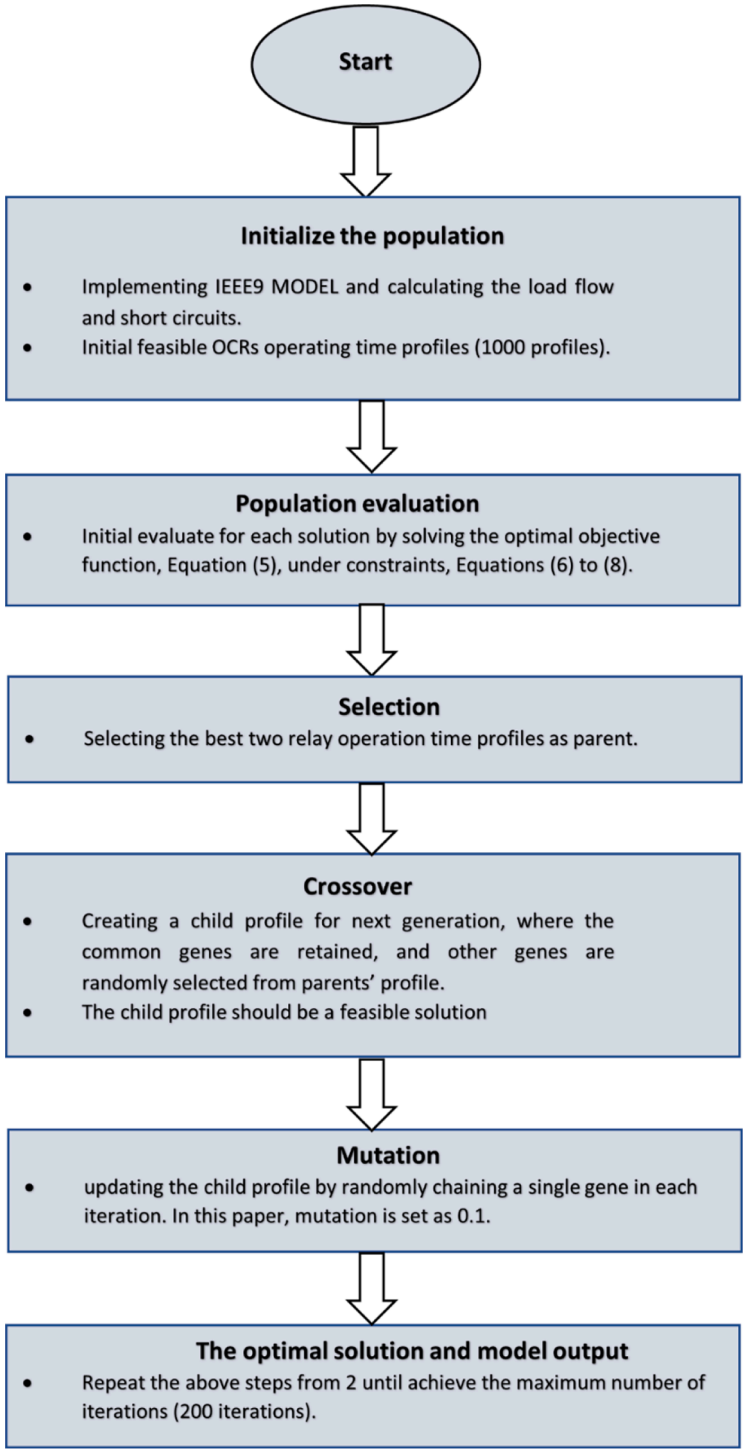


Fig. 8. A flow chart of GA for the OCR coordination problem.

and nonstandard characteristics with the new adaptive characteristics is represented as shown in Fig. 7 for various fault types. For instance, the fault will be within the maximum fault level if it is close to the power source and will be within the minimum fault level if it is at the end of the protected zone. The minimum operation time for the highest fault current must be guaranteed by the OC coordination mechanism.

As seen in Fig. 7, when comparing the nonstandard and standard curves with the proposed new adaptive protection, it is observed that the new approach offers a shorter operating time at point (B) (maximum fault). The minimal fault current for point (A) will result in a longer OC relay tripping time. However, the proposed new adaptive curve will continue to offer a shorter operational time compared to the other two curves. Minimal fault current conditions will normally occur more frequently in modern DN systems than in networks without DGs. To minimize the tripping time of OCRs and ensure that there are no miscoordination events, the suggested new adaptive curve will not be a fixed curve with a constant setting, it will be rather modified based on the voltage measurement. The following subsections will presnt the optimization problem for the OCRs in modern DNs based on nonstandard current-voltage characteristic and the optimization solver.

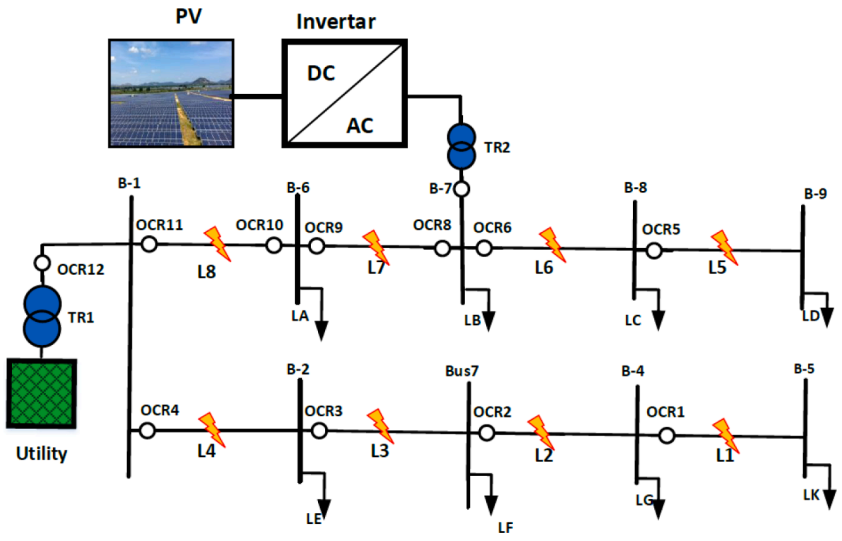


Fig. 9. IEEE 9 bus microgrid system.

3.2. Coordination problem formulation

This study formulates the OCRs coordination problem in DNs with DG as an optimization problem. The objective here is to identify the optimal TMS value that reduces the overall operational times for the OCRs while maintaining the selectivity between the primary and backup relays. In order to tackle the OCRs coordination problem and enhance the performance of the standard and the nonstandard curves for OCRs in modern DNs, the mathematical formulation of the proposed optimization problem and methodology are presented in this section.

3.2.1. Objective function

The TMS controls the operating time of the relays, as described in Eqs. (2), (4) and (5), therefore it is a primary factor to develop the OCR's coordination problem. In order to minimize the total operating time of the backup and primary relays, the objective function (O_f) is applied as described in Eq. (6). The O_f is the total operation time, t, for the relay for the total number of relays, N, and the total number of fault locations, K.

$$O_f = \sum_{n=1}^{N} \sum_{k=1}^{K} t_{n,k} \tag{6}$$

In addition, the O_f is restricted to the number of constraints, selectivity, and boundary constraints. First, in order to introduce a propositional operation time delay between the backup and primary backup OCRs, the selectivity and coordination constraint need to be applied. This constraint will help to reduce the network's power outage; the backup OCRs shouldn't function unless the primary OCRs operate. Based on the coordination time interval (CTI), the selectivity and coordination are formulated as a set of inequality constraints, as described in Eq. (7).

$$t_{back} - t_{prim} \geq CTI \tag{7}$$

where t_{back} and t_{prim} are the operating times for the backup and primary relays respectively. To achieve selectivity, the CTI is often chosen between 0.2 and 0.5 sbased on IEEE-242 [18,24]. In this paper, the CTI is set to 0.3 s. The second constraint criterion is the boundary for the OCR operating time and the relay setting. These constraints describe the minimum and maximum OCR operational times to sustain the operational time limitations, as presented in Eq. (8). The TMS is also required to be in a suitable margin as described in Eq. (9). However, the protective relays must act as rapidly and efficiently as possible; if the OCRs operate more slowly, the equipment would suffer different damages and the power supply will become unstable. The followings can be given as the minimum and maximum operational time bounds:

$$t_{min} \leq t_n \leq t_{max} \tag{8}$$

$$TMS_{min} \leq TMS_n \leq TMS_{max} \tag{9}$$

where t_{min} and t_{max} are the minimum and maximum operational times, respectively, for relay (n) with time operational (t_n) when the relay acting as the primary. The TMS of relay n is TMS_n and it will be between the minimum TMS, TMS_{min} , and the maximum TMS, TMS_{max} [17–19]. Finally, the PSM boundaries in this paper will range from 1.1 to 100 as proposed in [20,23,25].

3.3. GA optimization

In the previous subsection, the O_f optimization problem described the OCRs coordination problem in modern DN. In this section, the GA is described and used to solve complex OCRs coordination optimization problems. In general, the GA optimization algorithm provides highly efficient performance in solving complex power engineering problems and achieving the best optimal solutions [26,27]. The literature [27,28] showed that the GA is a powerful optimization algorithm in solving power flow problems for power systems with DG. Furthermore, GA has been wildly employed in solving relay coordination problems and different power protection applications [28]. Typically, the GA is simulated with a fixed population size, where the population represents potential answers (TMS) for the OCRs coordination problem. In the next stage, the population will be evaluated using an objective function called a fitness function as seen in Eq. (6). Then, using the techniques of selection, crossover, and mutation, a new population will be reproduced and evaluated to achieve the global solutions.

Fig. 8 presents the outline scheme of the GA in solving an OCRs coordination problem using standard and nonstandard curves. A set of initial population for the TMS and calculating the OCR operating time profiles is presented as seen using Eq. (5). Then the TMS and OCR's operating time profiles will be evaluating by the objective function seen in Eq. (6). The parents of the future generation are chosen in this paper as the best possible solution (best OCRs operating time performance). The parents will be used in the reproduction step by choosing the parent profiles for crossover. The common genes of the parents are retained for creating a new child of the TMS and OCRs operation time profiles, while the remaining genes are chosen at random from the parents. A feasibility test is used to determine whether or not the constraints of the child profile satisfies the power network or relay. If the child solution is rejected, where it is located in the infeasible region, another child profile

Table 2
Short-circuit fault currents and plug setting multiplier (PSM) for several LLL, LL, LG, LLG faults (Scenario 1).

Location of fault	LLL		LL		LG R1=0		LG R2=3		LLG R1=0		LLG R2=3		Relays
	KA	PSM	KA	PSM	KA	PSM	KA	PSM	KA	PSM	KA	PSM	
F1	4.380	38.8	4.047	35.8	4.944	43.8	1.979	17.5	5.250	46.5	1.161	10.2	OCR 1
	4.378	19.3	4.047	17.9	4.944	21.8	1.979	8.7	5.250	23.2	1.161	5.1	OCR 2
F2	4.882	21.6	4.641	20.5	5.710	25.2	2.081	9.2	6.113	27.0	1.192	5.2	OCR 2
	4.882	14.3	4.641	13.6	5.710	16.8	2.081	6.1	6.113	18.0	1.192	3.5	OCR 3
F3	5.496	16.1	5.479	16.1	6.810	20.	2.192	6.4	7.376	21.7	1.224	3.6	OCR 3
	5.496	12.1	5.479	12.0	6.810	15	2.192	4.8	7.376	16.2	1.224	2.6	OCR 4
F4	6.246	13.7	6.784	14.9	8.575	18.8	2.315	5.09	9.473	20.8	1.258	2.7	OCR 4
	6.246	9.13	6.784	9.9	8.575	12.5	2.315	3.3	9.473	13.8	1.258	1.8	OCR12
F5	4.380	38.8	4.047	38.8	4.944	38.8	1.979	17.5	5.250	38.8	1.161	10.2	OCR 5
	4.378	19.37	4.047	17.9	4.944	21.8	1.979	8.7	5.250	23.2	1.161	5.1	OCR 6
F6	4.882	21.60	4.641	20.5	5.710	25.2	2.081	9.20	6.113	27.0	1.192	5.2	OCR 6
	4.882	14.3	4.641	13.6	5.710	16.8	2.081	6.1	6.113	18.0	1.192	3.5	OCR 7
F7	5.496	16.1	5.479	16.1	6.810	20	2.192	6.4	7.376	21.7	1.224	3.6	OCR 7
	5.496	12.10	5.479	12.0	6.810	15	2.192	4.8	7.376	16.2	1.224	2.6	OCR 8
F8	6.246	13.7	6.784	14.9	8.575	18.8	2.315	5.0	9.473	20.8	1.258	2.7	OCR 8
	6.246	9.131	6.784	9.9	8.575	12.53	2.315	3.3	9.473	13.8	1.258	1.8	OCR 12

will be created by randomly reselecting the non-common genes until the child solution is located in the feasible region. The parent and child profiles will reach closer to one another with each iteration as the search moves closer to the optimal operating time profile for the OCRs coordination problem. Until the proposed threshold is achieved or the maximum number of iterations is reached, this process will be repeated.

4. Simulation results and discussion

The proposed novel adaptive protection scheme is evaluated and tested using two scale of common distribution systems. The two benchmarks distribution networks are equipped with DGs (two PV plants) to show the performance of proposed novel adaptive protection scheme compared to traditional coordination methods from under different fault scenarios and grid operation modes, as follows:

- IEEE 9-bus distribution network with two PV plants.
- IEEE 33-bus system, including PV plants.

4.1. IEEE-9 bus distribution network model

The suggested new adaptive voltage-current OCR coordination scheme is tested on an IEEE 9-bus test system, to find the optimal OCRs coordination and achieve a minimal total operational time. The power network system, as shown in Fig. 9, was created as a benchmark grid using the Canadian Urban with a four-bus feeder distribution system as the basis for a segment of two feeders [29]. This network is supplied by a main utility source that has an X/R ratio of 6, a short-circuit capacity of 500 MVA, and a line length of 500 m. In addition, the power network includes a 20 MVA, 115kV/12.47kV substation transformer with a 10% leakage reactance connecting the utility to the system. The derailed network specifications and parameters are presented in [29,30], where the network has four synchronous DG units (PV) with a combined rating of 5MVA and a subtransient reactance of 9.67%. A 480V/33 kV set-up transformer with 5% subtransient reactance is used to link each DG unit. The power system, as shown in Fig. 9, is protected by 12 OCRs with two-directional OCRs (OCR8 and OCR10) to block the PV system from increasing fault.

In this paper, different fault scenarios (from F1 to F8) are considered to test the performance of the proposed protection system. In Fig. 9, faults F1 and F8, represent the near-end and far-end fault locations on each line. In order to test and investigate the protection sensitivity and selectivity, two primary OCRs are assigned for each fault location, in addition to having a backup OCR for each primary OCR. For instance, in the event of a fault (i.e. F2), the primary relay will be operated by the OCR2, and the backup relay by the OCR3. The load flow was used to establish the CTR and set up the initial PS for each OCR. Additionally, a short-current calculation has been determined according to IEC-60909 for various operational circumstances and fault scenarios in various sections [3,10,29]. Fig. 9 presents the main OCR parameters: Plug Setting (PS) and Current Transformer Ratio (CTR) for each OCR. In this article, ETAP software was used to examine the proposed protection scheme and the short circuit.

The novel adaptive voltage-current OCR coordination strategy proposed is tested and reported in this section under different scenarios of the faults and power networks model. In this section, depending on the following configurations, we will assess the performance of the proposed OCR coordinating systems for power network models using GA:

- Scenario 1: The primary utility feeder is only fed from the electricity network grid. Here, the purpose is to investigate how the proposed OCR coordination systems perform on a conventional power network without DGs.
- Scenario 2: The PV system equipped to the electricity network grid as DG. In this case, achieving the ideal OCR coordination becomes more complex and difficult, as described in Sections 1 and 2. In this scenario, the suggested voltage-current scheme is evaluated for a power network connected to DGs and compared to standard and nonstandard characteristics presented in the literature.

4.1.1. Scenario 1: traditional power network

In order to assess the proposed adaptive OCR scheme for Scenario 1, various faults are simulated including different faults and short-circuit currents. Three-Phase Fault (LLL), Line to Line Fault (LL), Line to Ground Fault (LG), and Double Line to Ground Fault (LLG), as shown in Table 2. In this section, two values of ground resistance (R1=0 Ohm and R2=3 Ohm) are simulated for the related transmission line faults (LG and LLG). In general, the primary OCRs should operate as quickly as possible; they are considered to firstly work and then backup OCRs. Therefore, the PSM for each OCR is determined according to the maximum load currents on the line. Additionally, different PSM settings for the OCRs are related to the fault circumstances, as presented in Table 2. As an illustration, the PSM for OCR 1 is 21.6 for the LLL fault at F2, while OCR 2 is set at 14.3 for the same fault scenario.

The tripping time of all OCR pairs (primary and backup) under different fault scenarios is presented in Table 2 for the proposed voltage-current characteristic (PSC) approach. The PSC is compared to traditional Standard Inverse Time (STC) technique and non-standard current characteristic (NSC) approach [29] in terms of providing minimum tripping time while maintaining appropriate CTI between the OCR pairs.

Table 3
The OCRs tripping time over the proposed fault locations.

Fault location	Primary / Backup Relays	STC	CTI	NSC	CTI	PSC	CTI
		Time (s)	(old)	Time (s)	(old)	Time (s)	(adaptive)
F1	OCR1	0.019	0.3s	0.08	0.22s	0.001	0.069s
	OCR 2	0.318		0.30		0.07	
F2	OCR 2	0.306	0.305s	0.28	0.3s	0.04	0.12s
	OCR 3	0.61		0.58		0.16	
F3	OCR 3	0.58	0.3s	0.53	0.3s	0.09	0.18s
	OCR 4	0.88		0.83		0.27	
F4	OCR 4	0.83	0.307s	0.78	0.3s	0.14	0.05s
	OCR 12	1.137		1.08		0.19	
F5	OCR 5	0.019	0.291s	0.008	0.292s	0.001	0.06s
	OCR 6	0.31		0.30		0.08	
F6	OCR 6	0.30	0.31s	0.28	0.3s	0.04	0.12s
	OCR 7	0.61		0.58		0.16	
F7	OCR 7	0.58	0.3s	0.53	0.3s	0.09	
	OCR 8	0.88		0.83		0.27	0.18s
F8	OCR 8	0.83	0.307s	0.78	0.3s	0.14	0.05s
	OCR 12	1.137		1.08		0.19	

Table 4
Short-circuit faults currents and Plug Setting Multiplier (PSM) for several LLL, LL, LG, LLG faults (Scenario 2).

Location of fault	LLL		LL		LG RF=0		LG RF=3		LLG RF=0		LLG RF=3		Relay
	KA	PSM	KA	PSM	KA	PSM	KA	PSM	KA	PSM	KA	PSM	
F1	4.369	38.7	3.985	35.3	5.003	44.3	1.973	17.4	5.490	48.6	1.160	10.2	OCR 1
	4.369	19.3	3.985	17.6	5.003	22.1	1.973	8.7	5.490	24.2	1.160	5.1	OCR 2
F2	4.880	21.5	4.547	20.1	5.771	25.5	2.072	9.1	6.412	28.3	1.190	5.2	OCR 2
	4.880	14.3	4.547	13.3	5.771	16.9	2.072	6.1	6.412	18.8	1.190	3.5	OCR 3
F3	5.508	16.2	5.322	15.6	6.855	20.1	2.182	6.4	7.733	22.7	1.222	3.5	OCR 3
	5.508	12.1	5.322	11.7	6.855	15	2.182	4.8	7.733	17	1.222	2.6	OCR 4
F4	6.285	13.8	6.491	14.2	8.554	18.8	2.302	5	9.963	21.9	1.256	2.7	OCR 4
	6.285	9.1	6.491	9.4	6.676	9.7	1.797	2.6	7.773	11.3	0.980	1.4	OCR 12
F5	4.456	38.8	4.034	38.8	5.305	38.8	2.008	17.8	6.155	38.8	1.181	10.4	OCR 5
	4.456	18.5	4.034	16.7	5.305	22	2.008	8.3	6.155	25.5	1.181	4.9	OCR 6
F6	4.994	22	4.606	20.3	6.176	27.3	2.110	9.3	7.366	32.5	1.213	5.3	OCR 6
	0.347	1.4	0.320	1.3	1.985	8.2	1.438	5.9	2.365	9.8	0.390	1.6	OCR 7
	4.848	14.2	4.471	13.1	4.207	12.3	0.678	1.9	5.016	14.7	0.826	2.4	OCR 9
F7	5.496	16.1	5.232	15.4	5.069	14.9	1.515	4.4	6.307	18.5	0.849	2.5	OCR 8
	5.496	16.1	5.232	15.4	5.069	14.9	1.515	4.4	6.307	18.5	0.849	2.5	OCR 9
	5.496	12.1	5.232	11.5	5.069	11.1	1.515	3.3	6.307	13.8	0.849	1.8	OCR 11
F8	6.246	13.7	6.377	14.	6.535	14.3	1.695	3.7	8.070	17.7	0.924	2	OCR 10
	0.390	1.6	0.399	1.6	2.417	10	0.627	2.6	2.984	12.3	0.342	1.4	OCR 8
	6.246	13.7	6.377	14.	5.535	12.1	1.695	3.7	8.070	17.7	0.924	2	OCR 11
	6.246	9.1	6.377	9.3	6.535	9.5	1.695	2.4	8.070	11.7	0.924	1.3	OCR 12

Table 5
OCRs tripping time over all fault conditions for Scenario 2.

Fault location	Primary / Backup Relays	STC Time(s)	CTI (old)	NSC Time(s)	CTI (old)	PSC Time(s)	CTI (adaptive)
F1	OCR1	0.018		0.008		0.001	0.079s
	OCR2	0.31	0.308s	0.30	0.292s	0.08	
F2	OCR2	0.30		0.28		0.04	0.13s
	OCR3	0.60	0.3s	0.58	0.3s	0.17	
F3	OCR3	0.57		0.53		0.08	0.19s
	OCR4	0.87	0.3s	0.83	0.3s	0.27	
F4	OCR4	0.83		0.77		0.14	0.06s
	OCR12	1.13	0.3s	1.07	0.3s	0.20	
F5	OCR5	0.018		0.008		0.001	0.079
	OCR6	0.32	0.302s	0.30	0.292s	0.08	
F6	OCR6	0.30	0.3s	0.26	0.31s	0.038	
	OCR7	1.18		0.85		0.12	0.11s
	OCR 9	0.60	0.58s	0.57	0.32	0.006	
F7	OCR 8	0.57	0.3s	0.52	0.3s	0.083	
	OCR 9	0.024		0.02		0.002	0.1s
	OCR 11	0.323	0.29s	0.32	0.3s	0.101	
F8	OCR 10	0.026	0.956s	0.022	0.87s	0.0002	
	OCR 8	0.970		0.827		1.32	——
	OCR 11	0.307		0.297		0.054	
	OCR 12	1.136	0.829s	1.080	0.783	0.194	

Table 6
The overall operational time for STC, NSC and PSC curves over different power network scenarios.

Relay NO	Scenario 1 TMS STC	TMS NSC	TMS PSC	Scenario 2 TMS STC	TMS NSC	TMS PSC
OCR1	0.0101	0.0101	0.0101	0.0100	0.0101	0.0101
OCR 2	0.1387	0.1710	0.1710	0.1387	0.1712	0.1712
OCR 3	0.2386	0.2643	0.2643	0.2368	0.2643	0.2643
OCR 4	0.3220	0.3451	0.3451	0.3209	0.3452	0.3452
OCR 5	0.0101	0.0101	0.0101	0.0100	0.0100	0.0100
OCR 6	0.1388	0.1710	0.1710	0.1387	0.1664	0.1664
OCR 7	0.2351	0.2647	0.2647	0.0621	0.1608	0.1608
OCR 8	0.3190	0.3450	0.3450	0.2367	0.2582	0.2582
OCR 9	0.3673	0.3841	0.3841	0.0101	0.0101	0.0101
OCR 10				0.0100	0.0100	0.0100
OCR 11				0.1182	0.1316	0.1316
OCR 12				0.3672	0.3840	0.3840
overall operational time	9.3519	8.8475	1.8	11.2824	10.3534	2.99

In Scenario 1, the proposed PSC outperformed the STC and NSC approaches, where the STC and NSC recorded a delay in the OCR tripping time compared to the PSC, as shown in Table 3. For instance, the PSC technique achieved the minimum tripping time for the primary and backup relays in the event of an LLL fault at F2 with 0.04 sand 0.16 s compared to 0.28 s and 0.58 s for the NSC approach respectively. As can be seen, when the PSC technique is used, all OCRs record the minimum tripping times. The application of the PSC technique, as shown in Table 3, demonstrates the adaptive CTI changes of all OCRs to provide an accurate CTI among all fault circumstances. It is important to note that the voltage bus is less than 1 per unit in the LLL fault. Therefore, the PSM for the majority of fault situations in this scenario is higher than 20. Additionally, when the PSC technique is utilized, the entire OCR tripping time satisfies all optimization constrains. On the other hand, when the STC and NSC coordination strategy are used, the sensitivity task is not always satisfied. As shown in Table 3, the sensitivity is adequately given when the PSC approach is used, demonstrating the superiority of the suggested approach.

4.1.2. Scenario 2 test results: power network equipped with DGs

As illustrated in Fig. 9, the ideal OCR configuration is chosen for a power network supplied by a utility feeder and a PV system (Scenario 2) in order to assess the PSC technique. Table 4 reports the fault current in various circumstances with two fault resistances (FR=0 Ohm, FR=3 Ohm). Additionally, Table 4 provides the acquired parameters for all OCRs. The OCRs have varying PSM settings based on the fault situations in order to obtain the best OCRs coordination. For instance, the PSM for R2 is 21.5 when the LLL failure occurs at F1, whereas R3 is set at 14.3. The STC, NSC, and PSC OCRs scheme approaches are evaluated under Scenario 2 with results presented as seen in Table 5. The LLL fault tripping time for the primary and backup OCR pairs for the PSC, NSC and STC characteristics under Scenario 2 are also presented in Table 5. The PSC has decreased the total tripping time while maintaining acceptable CTI between the primary and backup OCR pairs compared to the traditional STC and NSC techniques, as shown in Table 5. For example, considering R2 and R3 during F2, the NSC method achieved 0.3 s and 0.6 s tripping times and the NSC achieved 0.28 s and 0.58 s, respectively. When comparing the STC and NSC to the tripping time under Scenario 2 for the PSC approach, the time was decreased to 0.04 s and 0.14 s for R2 and R3, respectively.

4.1.3. Discussion and comparison

The previous section delt with evaluating the performance of the suggested PSC strategy compared to traditional STC and non-standard current NSC approaches under two different scenarios. In general, Table 6 displays the overall tripping time for each OCR in Scenario 1 and Scenario 2. The final TMS settings and total operational time (tripping time) are displayed in Table 6. The generated settings for each coordination scheme (STC, NSC and PSC) are employed to determine the overall operational time and CTI using the ETAP software. Table 6 shows that the proposed PSC strategy outperformed the STC and NSC in term of total operational time of all OCRs for all scenarios. For instance, the total operational times of all OCRs in Scenario 2 for the STC and NSC approaches were 11.2824 s and 10.3534 s, respectively, while the PSC strategy achieve 2.99 s.

4.1.4. OCRs sensitivity based on high fault resistance (HFR)

In this section, the effectiveness of PSC, NSC, and STC is evaluated by taking into account various imbalance fault types and ground fault

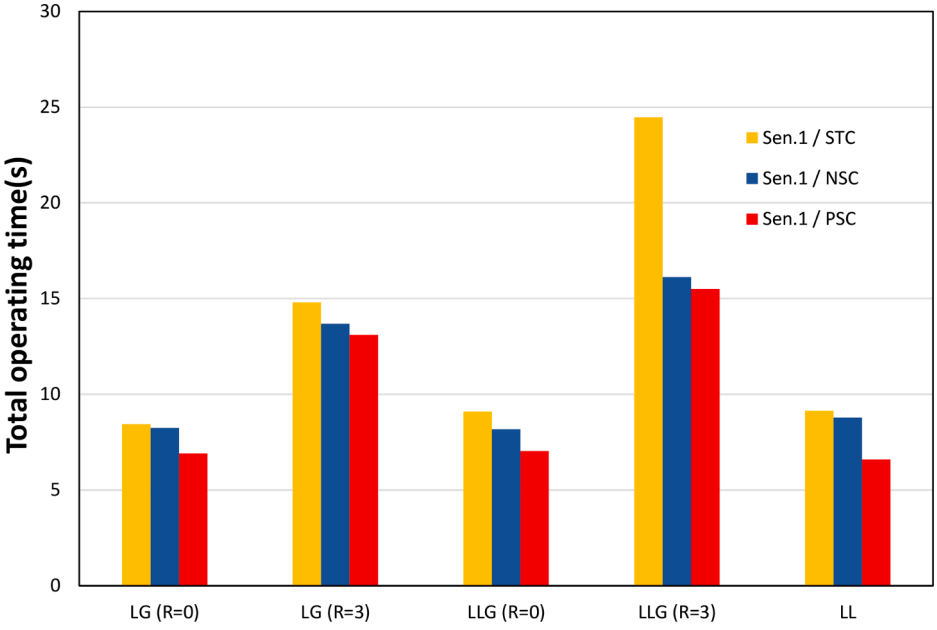


Fig. 10. The STC, NSC and PSC simulation results over a different unsymmetrical fault types in Scenario 1.

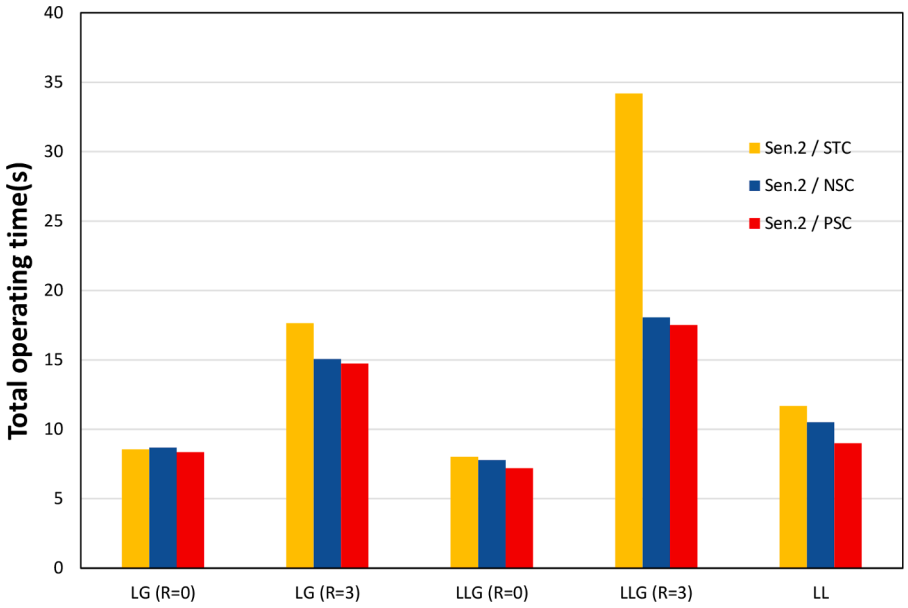


Fig. 11. The STC, NSC and PSC simulation results over a different unsymmetrical fault types in Scenario 2.

Table 7
Optimal overall operational time for STC, NSC and PSC curves over different power network scenarios.

Relay NO	Scenario 1 TMS	TMS	TMS	Scenario 2 TMS	TMS	TMS
	STC	NSC	PSC	STC	NSC	PSC
OCR1	0.0101	0.0101	0.0102	0.0100	0.0101	0.0106
OCR 2	0.1387	0.1710	0.67	0.1387	0.1712	0.6416
OCR 3	0.2386	0.2643	0.73	0.2368	0.2643	0.7081
OCR 4	0.3220	0.3451	0.69	0.3209	0.3452	0.6725
OCR 5	0.0101	0.0101	0.01	0.0100	0.0100	0.0125
OCR 6	0.1388	0.1710	0.667	0.1387	0.1664	0.6142
OCR 7	0.2351	0.2647	0.73	0.0621	0.1608	0.5703
OCR 8	0.3190	0.3450	0.697	0.2367	0.2582	0.0586
OCR 9	0.3673	0.3841	1.15	0.0101	0.0101	0.0103
OCR 10				0.0100	0.0100	0.0108
OCR 11				0.1182	0.1316	0.3926
OCR 12				0.3672	0.3840	1.115
overall operational time (Seconds)	9.3519	8.8475	5.1	11.2824	10.3534	4.74

impedances for Scenarios 1 and 2. Figs. 10 and 11 present the results of STC, NSC and PSC over a different unsymmetrical fault. As can be seen, the PSC reduced the total operational time and outperformed the traditional approach (STC) and non-standard current (NSC). For instance, the total operating time for the PSC approach in Scenario 1 and Scenario 2 during the LG fault condition were 6.9 s and 8.36 s, respectively, while for the NSC were 8.23 s and 8.69 s respectively, and for the STC approach were 8.42 s and 8.640 s respectively as well. In general, the PSC approach was able to deal with different fault scenario conditions and presented a strong and superior protection system with a quick response time. The proposed PSC handled the different fault resistance with high selectivity performance and without any records of miscoordination events

4.1.5. *Optimal coordination for voltage-current characteristics*

This section provides a comparison between the proposed voltage-current characteristic (PSC), non-standard characteristic NSC, and standard characteristic STC over the two scenarios in this paper with a fixed CTI that is equal to 0.3 s. Generally, the total tripping time for all OCRs in Scenarios 1 and 2 are shown in Table 7. In addition, the obtained OCRs settings for each scenario are also presented in Table 7, which were computed using GA optimization technique. These optimal settings are then used to coordinate the OCRs in the optimal case and are

Table 8
Optimal OCRs tripping time over all fault conditions for Scenario 1.

Fault location	Primary / Backup Relays	STC	CTI	NSC	CTI	PSC	CTI
		Time (s)		Time (s)		Time (s)	(Fixed)
F1	OCR1	0.019	0.3s	0.08	0.22s	0.001	0.3s
	OCR 2	0.318		0.30		0.3	
F2	OCR 2	0.306	0.305s	0.28	0.3s	0.155	0.3s
	OCR 3	0.61		0.58		0.45	
F3	OCR 3	0.58	0.3s	0.53	0.3s	0.236	0.3s
	OCR 4	0.88		0.83		0.532	
F4	OCR 4	0.83	0.307s	0.78	0.3s	0.28	0.3s
	OCR 12	1.137		1.08		0.583	
F5	OCR 5	0.019	0.291s	0.008	0.292s	0.001	0.3s
	OCR 6	0.31		0.30		0.03	
F6	OCR 6	0.30	0.31s	0.28	0.3s	0.155	0.3s
	OCR 7	0.61		0.58		0.452	
F7	OCR 7	0.58	0.3s	0.53	0.3s	0.236	
	OCR 8	0.88		0.83		0.537	0.3s
F8	OCR 8	0.83	0.307s	0.78	0.3s	0.58	0.3s
	OCR 12	1.137		1.08		0.28	

Table 9
Optimal OCRs tripping time over all fault conditions for Scenario 2.

Fault location	Primary / Backup Relays	STC Time(s)	CTI	NSC Time(s)	CTI	PSC Time(s)	CTI (Fixed)
F1	OCR1	0.018		0.008		0.001	0.292s
	OCR2	0.31	0.308s	0.30	0.292s	0.293	
F2	OCR2	0.30		0.28		0.15	0.3s
	OCR3	0.60	0.3s	0.58	0.3s	0.44	
F3	OCR3	0.57		0.53		0.23	0.3s
	OCR4	0.87	0.3s	0.83	0.3s	0.52	
F4	OCR4	0.83		0.77		0.27	0.3s
	OCR12	1.13	0.3s	1.07	0.3s	0.58	
F5	OCR5	0.018		0.008		0.001	0.3s
	OCR6	0.32	0.302s	0.30	0.292s	0.301	
F6	OCR6	0.30	0.3s	0.26	0.31s	0.146	
	OCR7	1.18		0.85		0.446	0.3s
	OCR 9	0.60	0.58s	0.57	0.32	0.006	
F7	OCR 8	0.57	0.3s	0.52	0.3s	0.01	
	OCR 9	0.024		0.02		0.003	0.3s
	OCR 11	0.323	0.29s	0.32	0.3s	0.302	
F8	OCR 10	0.026	0.956s	0.022	0.805s	0.00002	
	OCR 8	0.970		0.827		0.299	0.3s
	OCR 11	0.307		0.297		0.16	
	OCR 12	1.136	0.829s	1.080	0.783	0.565	

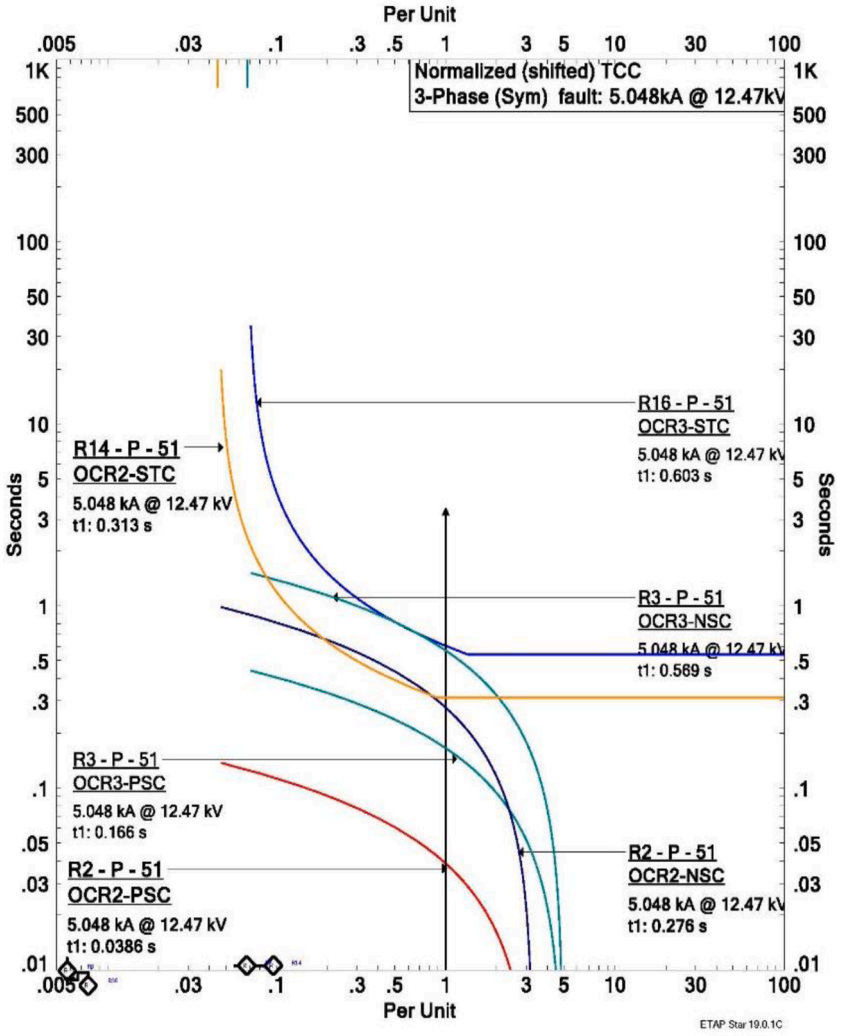


Fig. 12. Characteristics curves of relays (OCR2) when a fault occurs at location F2 for Scenario 1.

utilized to examine their performance in terms of the overall operational time and CTI as shown in Tables 8 and 9. Firstly, the PSC approach was found to outperform the STC and NSC in terms of minimum tripping time. For example, the PSC approach reduces the overall operational time of all OCRs compared to STC and NSC by 57.9% and 54.2% in Scenario 2, respectively. The overall operational times of all OCRs in

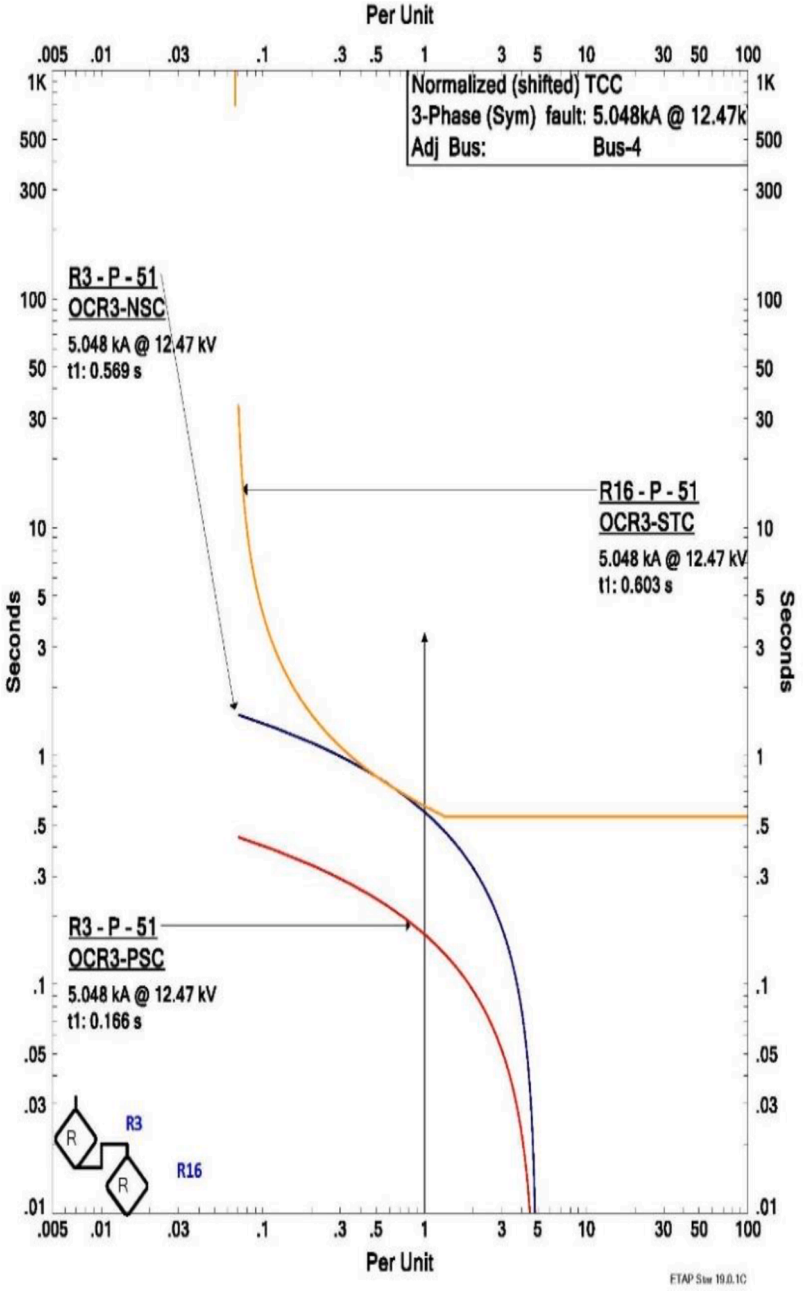


Fig. 13. Characteristics curves of relays (OCR3) when a fault occurs at location F2 for Scenario 1.

Scenario 1 were 9.3519 s ,8.8475 s and 5.1 s for STC, NSC and PSC approaches, respectively.

4.1.6. Design evaluation using (ETAP)

ETAP software is one of the most helpful programs for understating and investigating relays performance. ETAP is a standardized computer program that employs protection and network models using an intuitive friendly interface. This software is used to investigate the effectiveness and viability of the strategy suggested in this paper. IT is employed in this work to acquire the best coordination of the OCR method without malfunctioning. In this instance, the distribution system of the four-bus Canadian Urban Benchmark portion (two feeders) is being examined. The performance of the PSC strategy in comparison to the NSC and STC approaches is assessed as seen in Figs. 12 to 14. For the two proposed scenarios in this paper (with and without DG), the simulation models were created to best coordinate the OCR protection mechanism. Figs. 12 to 15 show the Time-Current Characteristics (TCC) for the OCR system using ETAP package, Pre-setting, and GA utilized for the coordination problem of overcurrent relay protection using manual calculations. The two situations that are suggested to replicate the OCR coordination between the primary and backup OCR pairs are:

- Firstly, Figs. 12 and 13 show OCRs (OCR2 and OCR3) coordinating with relays (OCR2 and OCR3), when a failure occurs at location F2 in Scenario 1. In a fault case, the primary OCR2 will operate first, followed by the backup the OCR3 in the event when the primary OCR2 is unable to work. As seen in Fig. 8, the OCR2 will run in 0.04 s and the OCR3 will operate in 0.16 s as a backup when the PSC technique is employed to coordinate the OCRs. The tripping time for the OCR2 was 0.28 s and 0.58 s for the OCR3, when using NSC is used, while the tripping times were 0.307 s and 0.611 s, respectively, when using

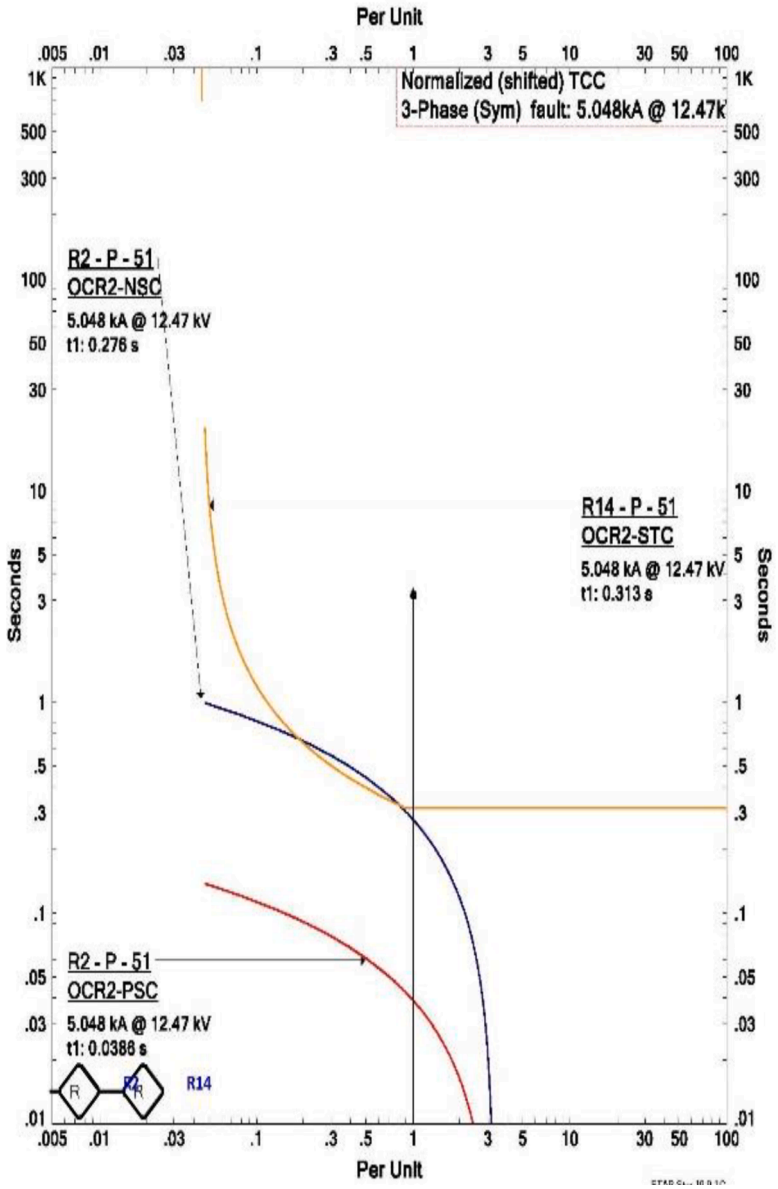


Fig. 14. Characteristics curves of relays (OCR2) when a fault occurs at location F2 for Scenario 2.

STC. However, the PSC approach provides the most value decrease in total operational time.

- Secondly: The relay coordination between OCR2 and OCR3 in the situation of fault F2 when the PV system is linked to the power network (Scenario 2) is shown in Figs. 14 and 15. Note that when using PSC, the tripping time for the primary OCR2 was 0.04 s, while the tripping time for the backup OCR3 was 0.17 s compared to 0.28 s and 0.58 s for NSC. Additionally, when STC is employed, the tripping time for primary the OCR2 was 0.307 s and for the backup OCR3 was 0.699 s. In Scenario 2 for instance, the suggested PSC technique resulted in shorter tripping times in all fault situations with a discernible rise in fault current level due to the contribution of the PV system. Overall, when PSC was used instead of NSC or STC, the tripping times of all OCRs in the two scenarios were significantly reduced. In general, the proposed PSC showed high selectivity performance without any records of miscoordination events.

4.2. *IEEE-33 bus distribution network model*

In order to investigate the transferability of the proposed PSC approach to other power systems, a larger network model based on IEEE-33 bus system. Fig. 16 presents the proposed IEEE-33 bus system equipped with two 4-MW PV plants. The data and information of the IEEE-33 bus system is common and available in [31,32]. In this work, the proposed IEEE-33 bus system including 32 OCRs are simulated system by using ETAP software. Table 10 shows the main OCRs settings including CTR and IP. The TMS limits for OCRs were chosen by solving the coordination problems in range of 0.01 and 3.

To evaluate the performance of the proposed PSC approach compared to STC and NSC approaches, three phase fault (LLL) for different locations are generated, as shown in Fig. 16. Table 11 displays the tripping time for each OCR over the different LLL fault scenarios. The PSC approach outperformed the STC and NSC in terms of minimum tripping time over all faults and without any miscoordination records as shown in Table 11. For example, the PSC approach reduces the operational time at F18 for primary relay (OCR18) and backup relay (OCR1) to 0.0117 and 0.0582 s, respectively, compared to 0.645 and 1.701 s for STC approach, and 0.586 and 1.165 s for NSC approach, respectively. The results of the overall operational time for all coordination approaches, as shown in Fig. 17, shows that the proposed PSC approach

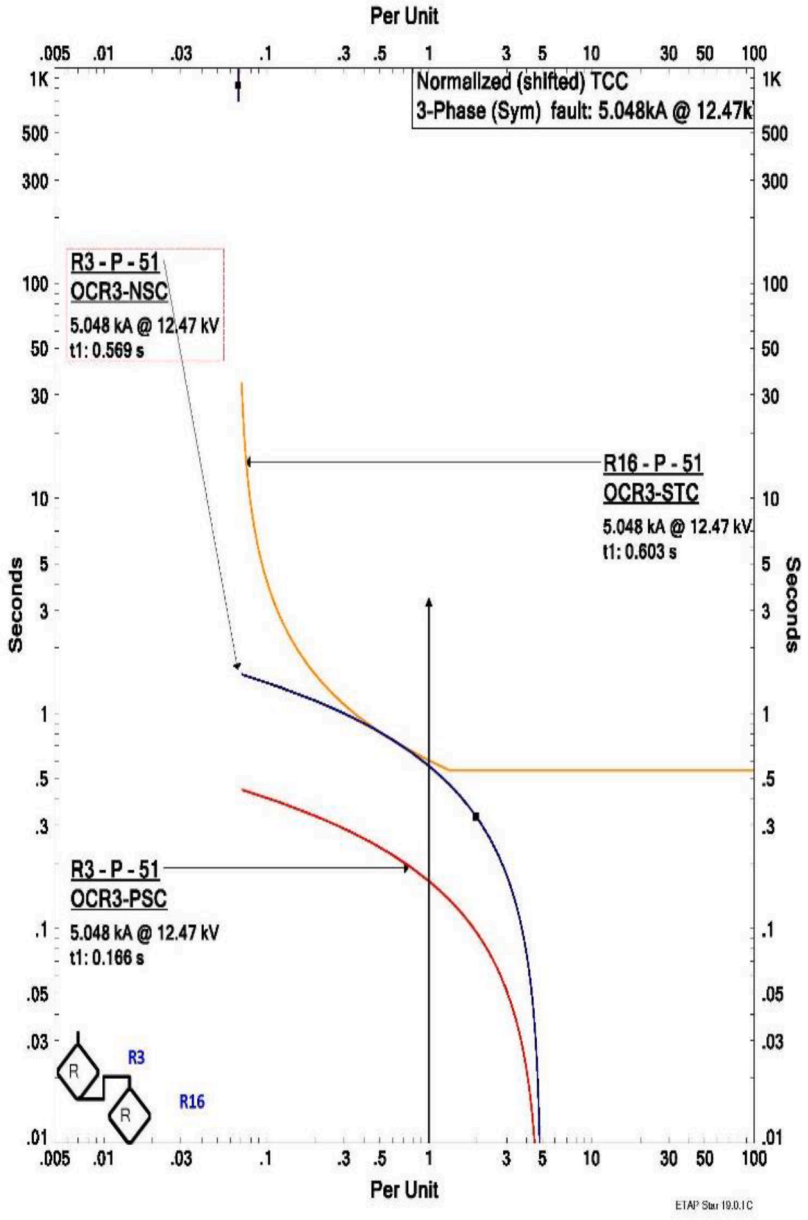


Fig. 15. Characteristics curves of relays (OCR3) when a fault occurs at location F2 for Scenario 2.

outperformed the NSC and STC approaches by reducing overall operational time by 96.4% and 96.9%, respectively.

5. Conclusions

The main objective of this paper was to integrate a fast and adaptive OCR coordination technique into an actual power network architecture (with and without DGs). The PSC technique has used to set all constraints related to the OCR coordination problem and achieve the minimum tripping time by solving the OCRs coordination problem using GA algorithm approach based on novel nonstandard current-voltage characteristic. Additionally, the research in this paper intends to propose a creative and simple nonstandard current-voltage characteristic that can be used in different power network ground fault scenarios. The suggested PSC, NSC, and STC techniques were developed and compared under fault scenarios (including various fault and impedances) for a power grid-connected (IEEE 9 and 33 bus) with and without DG or islanding operating mode. Results of the suggested novel PSC technique showed that the proposed adaptive OCR approach (PSC) met the coordinating assignment without any miscoordination records, and also with a notable reduction in the overall operational time. For instance, the total operational times of all OCRs in Scenario 2 for the STC and NSC approaches were 11.2824 s and 10.3534 s, respectively, while the PSC strategy achieve 2.99 s.

Authorship statement

All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be submitted to or published in any other publication before its appearance in Electric Power Systems Research.

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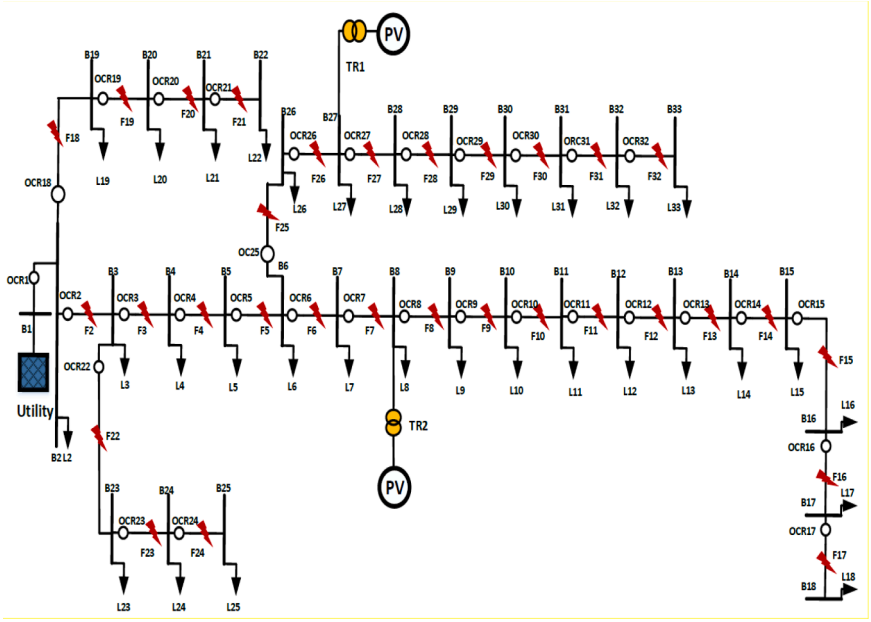


Fig. 16. IEEE-33 bus microgrid system.

Table 10
OCRs setting for the proposed IEEE-33 bus system.

Relay	CTR	IP
OCR1 and OCR2	200/1	50
OC3 to OC32	100/1	50

that all authors have participated in drafting the article or revising it critically for important intellectual content; approval of the final version.

CRediT authorship contribution statement

Eyad Zarour: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feras Alasali:** Conceptualization, Methodology,

Table 11
Optimal OCRs tripping time over different LLL fault locations for IEEE-33 bus microgrid system.

Location of fault	Relays	LLL (A)	STC (sec)	NSC (sec)	PSC (sec)	Location of fault	Relays	LLL (A)	STC (sec)	NSC (sec)	PSC (sec)
F2	OCR2	316	3.354	2.979	0.0595	F18	OCR18	464	0.645	0.586	0.0117
	OCR1	316	3.578	3.178	0.1589		OCR1	150	1.701	1.165	0.0582
F3	OCR3	302	3.211	2.832	0.0566	F19	OCR19	445	0.438	0.398	0.0079
	OCR2	302	3.440	3.034	0.1517		OCR18	445	0.657	0.598	0.0299
F4	OCR4	291	3.045	2.669	0.0533	F20	OCR20	426	0.223	0.203	0.0040
	OCR3	291	3.279	2.874	0.1437		OCR19	426	0.447	0.407	0.0203
F5	OCR5	261	2.916	2.498	0.0499	F21	OCR21	409	0.0326	0.029	0.0005
	OCR4	261	3.249	2.783	0.1391		OCR20	409	0.228	0.207	0.010
F6	OCR6	281	2.670	2.324	0.0464	F22	OCR22	440	0.4408	0.401	0.0080
	OCR5	281	2.789	2.428	0.1214		OCR2	286	2.445	2.136	0.1068
F7	OCR7	310	2.260	2.002	0.0400	F23	OCR23	440	0.220	0.201	0.004
	OCR6	165	3.881	2.806	0.0561		OCR22	440	0.4408	0.401	0.0200
F8	OCR8	415	1.845	1.677	0.0335	F24	OCR24	422	0.0321	0.029	0.0005
	OCR7	194	3.055	2.381	0.1190		OCR23	422	0.224	0.204	0.0102
F9	OCR9	395	1.658	1.504	0.0300	F25	OCR25	347	1.594	1.433	0.0286
	OCR8	395	1.890	1.715	0.0857		OCR5	256	2.234	1.905	0.0952
F10	OCR10	390	1.435	1.301	0.0260	F26	OCR26	424	1.217	1.1073	0.0221
	OCR9	390	1.669	1.513	0.0756		OCR25	424	1.442	1.311	0.0655
F11	OCR11	387	1.206	1.093	0.0218	F27	OCR27	423	1.058	0.962	0.0192
	OCR10	387	1.440	1.306	0.0653		OCR26	423	1.219	1.108	0.0554
F12	OCR12	383	1.011	0.915	0.0183	F28	OCR28	407	0.849	0.772	0.0154
	OCR11	383	1.212	1.098	0.0549		OCR27	407	1.078	0.979	0.0489
F13	OCR13	355	0.840	0.756	0.0151	F29	OCR29	392	0.632	0.573	0.0114
	OCR12	355	1.050	0.946	0.0473		OCR28	392	0.865	0.785	0.039
F14	OCR14	343	0.641	0.576	0.0115	F30	OCR30	378	0.406	0.368	0.0073
	OC13	343	0.855	0.768	0.0384		OCR29	378	0.644	0.583	0.0291
F15	OCR15	334	0.433	0.388	0.0077	F31	OCR31	364	0.207	0.187	0.0037
	OCR14	334	0.650	0.582	0.0291		OCR30	364	0.414	0.374	0.0187
F16	OCR16	323	0.220	0.196	0.0039	F32	OCR32	353	0.035	0.0316	0.0006
	OC15	323	0.441	0.393	0.0196		OC31	353	0.2107	0.189	0.0094
F17	OC17	300	0.038	0.033	0.0006						
	OC16	300	0.230	0.202	0.010						

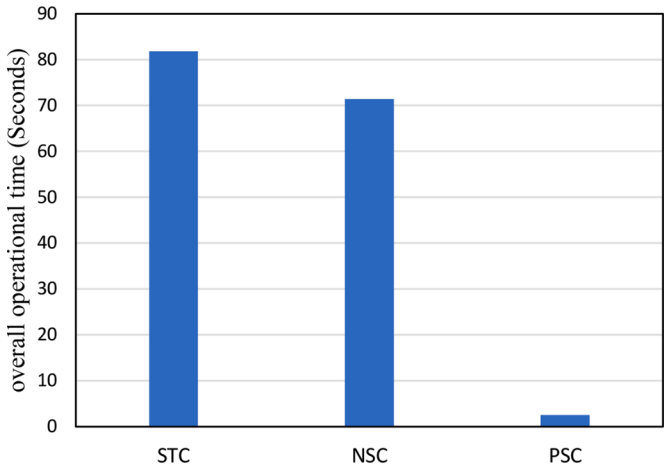


Fig. 17. The STC, NSC and PSC simulation results over a different LLL fault location for IEEE-33 bus network.

Validation, Resources, Writing – review & editing, Supervision. **Othman Alsmadi**: Methodology,Validation, Resources, Writing – review & editing, Supervision. **Naser El-Naily**: Validation, Formal analysis, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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